

Derived representation schemes and noncommutative formal smoothness

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Noncommutative algebraic geometry

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Algebraic geometry	Noncommutative algebraic geometry model
CommAlg _{k}	Alg _{k}

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Definition/Lemma [CQ95, KR00]

An associative algebra A is called **formally smooth** (or quasi-free) if any of the following equivalent conditions hold:

- 1 A has cohomological dimension ≤ 1 with respect to Hochschild cohomology.
- 2 The universal bimodule of derivations $\Omega_{\text{nc}}^1(A)$ is a projective bimodule.
- 3 A satisfies the lifting property with respect to nilpotent extensions in $\mathbf{Alg}_k$.

We also say A **smooth** if A is furthermore finitely generated.

Back to the table

Algebraic geometry	Noncommutative algebraic geometry model
CommAlg_k Spec A Proj A	Alg_k Mod – A (Mod – $A/\mathbf{Tor}$, s , $\mathcal{O}$)
Complete intersection Smooth	Noncommutative complete intersection Associative smoothness

Question

How to justify the noncommutative algebraic geometry model / properties?

Definition/Lemma

Fix an associative algebra A . All its representations on a finite-dimensional k -vector space V form an affine scheme called the **representation scheme**, denoted by $\text{Rep}_V(A)$.

Kontesvich-Rosenberg Principle

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The principle

Geometrically meaningful properties of A should induce corresponding classical geometric properties on $\text{Rep}_V(A)$.

Kontsevich-Rosenberg Principle: An example

Theorem [Gin05]

Fix an associative algebra A . If A is formally smooth, then $\mathrm{Rep}_V(A)$ is a formally smooth affine scheme.

Problem

It is generally difficult to study $\text{Rep}_V(A)$. Usually they are very singular, having very large dimension, etc.

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Idea to improve

Involving derived algebraic geometry.

Deriving the representation scheme

Theorem [BKR13]

Fix an finite-dimensional (graded) vector space V . For any DG algebra A , the derived representation scheme $\mathrm{DRep}_V(A)$ exists.

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More precisely, varying A , there is a Quillen pair

$$\mathcal{O}(\mathrm{Rep}_V(-)) : \mathbf{DGA}_k \rightleftarrows \mathbf{CDGA}_k : \underline{\mathrm{End}}(V) \otimes -$$

so that the left derived functor $\mathcal{O}(\mathrm{DRep}_V(-))$ of $\mathcal{O}(\mathrm{Rep}_V(-))$ exists.

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Definition

The homology groups^a

$$H_*(\mathcal{O}(\mathrm{DRep}_V(A)))$$

are called the **representation homology**, denoted by $\mathrm{HR}_*(A, V)$.

^aWe will always use homological convention.

Remark

One can think $\mathcal{O}(\mathrm{Rep}_V(-))$ is a “tensor product” and the representation homology are similar to Tor.

Deriving the Kontsevich-Rosenberg Principle

Theorem [BFR14]

Fix an associative algebra A .

If A is formally smooth, and admits a semifree resolution in $\mathbf{DGA}_{\geq 0}^k$ which is finitely generated in each degree,

then $\mathrm{DRep}_V(A)$ is quasi-isomorphic to $\mathrm{Rep}_V(A)$.

Back to derived algebraic geometry

In derived algebraic geometry, general spaces are understood to be approximated by well-behaved ones. The theorem suggests that noncommutative smooth spaces are already “well-behaved,” in the sense that no higher homological information is needed for such approximations—these spaces are approximated by themselves.

Failure to be a full characterization

Examples

The quantum plane $k[x, y]_q := \frac{k\langle x, y \rangle}{xy - qyx}$ is not formally smooth when q is not a root of unity, but $\mathrm{DRep}_V(k[x, y]_q) \simeq \mathrm{Rep}_V(k[x, y]_q)$ for all V . (See [BFR14])

Idea to remedy it: More choices of coefficients

Back to the definition of representations

A representation of an associative algebra A is defined to be an algebra homomorphism

$$A \rightarrow \text{End}_k(V)$$

for a finite-dimensional vector space V over k .

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More generally, one may replace $\text{End}_k(V)$ by a finite-dimensional algebra M and consider algebra homomorphisms

$$A \rightarrow M$$

which we call **representations of A with coefficient M** .

(Derived) Representation scheme with coefficient

Theorem [BFP⁺17]

Fix an finite-dimensional algebra M .

For any algebra A , all its representations with coefficient in M form an affine scheme $\text{Rep}_M(A)$.

Furthermore, the derived representation scheme $\text{DRep}_M(A)$ exists for any DG algebra A .

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Smoothness test

Theorem [L], proof very similar to [Gin05]

Fix an associative algebra A . If A is formally smooth, then $\mathrm{Rep}_M(A)$ is a formally smooth affine scheme.

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Fix an associative algebra A .

If A is formally smooth, and admits a semifree resolution in $\mathbf{DGA}_k^{\geq 0}$ which is finitely generated in each degree,

then $\mathrm{DRep}_V(A)$ is quasi-isomorphic to $\mathrm{Rep}_V(A)$.

Example: $k[x, y]_q = \frac{k\langle x, y \rangle}{\langle xy - qyx \rangle}$, q not a root of unity. Choose $M = (\mathfrak{n}_3)_+$ (strictly upper triangular 3×3 matrices, augmented).

```

0
1
2
q8 : Complex
i9 : prune HH C

q9 = cokernel ( sy vx - uy + xy + 2sz sx s^2 )  $\xleftarrow{0}$  cokernel  $\begin{pmatrix} \begin{matrix} \{3\} \\ \{3\} \\ \{3\} \\ \{3\} \end{matrix} & \begin{pmatrix} s & 0 & 2z & 0 & -y & v+y & 0 \\ 0 & s & u & v+y & 0 & 0 & 0 \\ 0 & 0 & -x & -y & 0 & s & 0 \\ 0 & 0 & 0 & 2z & x & -u & s \end{pmatrix} \end{pmatrix} \xleftarrow{0} \text{cokernel } \{5\} (s vx - uy + xy)$ 
0
2

q9 : Complex
i10 : D = M_3*N_3 - q*N_3*M_3
q10 =  $\begin{pmatrix} \frac{1}{2}st & \frac{1}{2}su + \frac{1}{2}tx & \frac{1}{2}sw + vx - \frac{1}{2}uy + \frac{1}{2}tz \\ 0 & \frac{1}{2}st & \frac{1}{2}sv + \frac{1}{2}ty \\ 0 & 0 & \frac{1}{2}st \end{pmatrix}$ 
q10 : Matrix A^3  $\leftarrow A^3$ 
i11 : C = koszulComplex {D_(0,0),D_(0,1),D_(0,2),D_(1,2)}
q11 =  $A^1 \xleftarrow{0} \begin{pmatrix} \frac{1}{2}st & \frac{1}{2}su + \frac{1}{2}tx & \frac{1}{2}sw + vx - \frac{1}{2}uy + \frac{1}{2}tz & \frac{1}{2}sv + \frac{1}{2}ty \end{pmatrix} A^4 \xleftarrow{1} \begin{pmatrix} -\frac{1}{2}su - \frac{1}{2}tx & -\frac{1}{2}sw - vx + \frac{1}{2}uy - \frac{1}{2}tz & 0 & -\frac{1}{2}sv - \frac{1}{2}ty & 0 & 0 \\ \frac{1}{2}st & 0 & -\frac{1}{2}sw - vx + \frac{1}{2}uy - \frac{1}{2}tz & 0 & -\frac{1}{2}sv - \frac{1}{2}ty & 0 \\ 0 & \frac{1}{2}st & \frac{1}{2}su + \frac{1}{2}tx & 0 & 0 & -\frac{1}{2}sv - \frac{1}{2}ty \\ 0 & 0 & 0 & \frac{1}{2}st & \frac{1}{2}su + \frac{1}{2}tx & \frac{1}{2}sw + vx - \frac{1}{2}uy + \frac{1}{2}tz \end{pmatrix} A^6 \xleftarrow{2} \begin{pmatrix} \frac{1}{2}sw + vx - \frac{1}{2}uy - \frac{1}{2}st \\ -\frac{1}{2}su - \frac{1}{2}tx \\ \frac{1}{2}st \\ 0 \\ 0 \\ 0 \end{pmatrix}$ 
0
1
2

q11 : Complex
i12 : prune HH C
q12 = cokernel ( sw + 2vx - uy + tz sv + ty su + tx st )  $\xleftarrow{0}$  cokernel {4} ( t s 2vx - uy )
0
1

q12 : Complex
i13 :
```

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```

i5 : M_3 = matrix{{s,x,z},{0,s,y},{0,0,s}}
o5 = 
$$\begin{pmatrix} s & x & z \\ 0 & s & y \\ 0 & 0 & s \end{pmatrix}$$

o5 : Matrix  $A^3 \leftarrow A^3$ 
i6 : N_3 = matrix{{t,u,w},{0,t,v},{0,0,t}}
o6 = 
$$\begin{pmatrix} t & u & w \\ 0 & t & v \\ 0 & 0 & t \end{pmatrix}$$

o6 : Matrix  $A^3 \leftarrow A^3$ 
i7 : -- D = M_3*N_3 - q*M_3*M_3
D = N_3*M_3 - M_3*N_3 - M_3*M_3
o7 = 
$$\begin{pmatrix} -s^2 & -2sx & -vx+uy-xy-2sz \\ 0 & -s^2 & -2sy \\ 0 & 0 & -s^2 \end{pmatrix}$$

o7 : Matrix  $A^3 \leftarrow A^3$ 
i8 : C = koszulComplex {D_(0,0),D_(0,1),D_(0,2),D_(1,2)}
o8 = 
$$A^1 \xleftarrow{0} \begin{pmatrix} -s^2 & -2sx & -vx+uy-xy-2sz & -2sy \\ 0 & -s^2 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} A^4 \xleftarrow{1} \begin{pmatrix} 2sx & vx-uy+xy+2sz & 0 & 2sy & 0 & 0 \\ 0 & 0 & vx-uy+xy+2sz & 0 & 2sy & 0 \\ 0 & -s^2 & -2sx & 0 & 0 & 2sy \\ 0 & 0 & 0 & -s^2 & -2sx & -vx+uy-xy-2sz \end{pmatrix} A^6 \xleftarrow{2} \begin{pmatrix} -vx+uy-xy-2sz & -2sy & 0 \\ 2sx & 0 & -2sy \\ -s^2 & 0 & 0 \\ 0 & 2sx & vx-uy+xy+2sz \\ 0 & -s^2 & 0 \\ 0 & 0 & -s^2 \end{pmatrix} A^9$$

o8 : Complex
i9 : prune HH C
o9 = cokernel  $\begin{pmatrix} sy & vx-uy+xy+2sz & sx & s^2 \end{pmatrix} \xleftarrow{0} \text{cokernel} \begin{pmatrix} \{3\} \begin{pmatrix} s & 0 & 2z & 0 & -y & v+y & 0 \\ 0 & s & u & v+y & 0 & 0 & 0 \\ 0 & 0 & -x & -y & 0 & s & 0 \\ 0 & 0 & 0 & 2z & x & -u & s \end{pmatrix} \end{pmatrix} \xleftarrow{0} \text{cokernel} \{5\} \begin{pmatrix} s & vx-uy+xy \end{pmatrix}$ 

```

Implication of the computations

Both the quantum plane and the Jordan plane are not smooth.

What can we expect?

Conjecture:

A associative algebra A over k which admits a semifree resolution in $\mathbf{DGA}_k^{\geq 0}$ which is finitely generated in each degree,

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A associative algebra A over k which admits a semifree resolution in $\mathbf{DGA}_k^{\geq 0}$ which is finitely generated in each degree, is smooth if and only if for **every** finite-dimensional algebras M ,

$$\mathrm{HR}_i^{\mathrm{Alg}}(A, M) = 0 \quad \text{for } i > 0$$

and $\mathrm{HR}_0^{\mathrm{Alg}}(A, M) \neq 0$.

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Implications:

- Provides a numerical characterization of associative smoothness.
- Converts a noncommutative problem into a (possibly more tractable) commutative one.
- A complete realization of the derived Kontsevich–Rosenberg principle for formal smoothness.

Can we formalize the KR-principle for nonaffine noncommutative spaces?
Or for other properties?

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Thank you!